

Code in Place 2025

Stanford CS106A

Section - Week 1

Welcome to Section!

David Tsai, Code in Place

Today's Agenda



Introductions

Icebreakers



Norms

Guidelines



Concept Review

Karel, Control Flow



Practice Problems

"Hospital Karel"

Please Turn On Your Camera



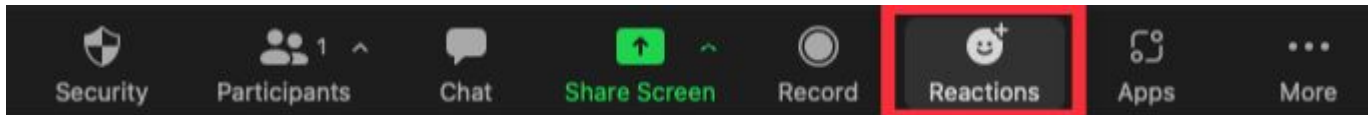
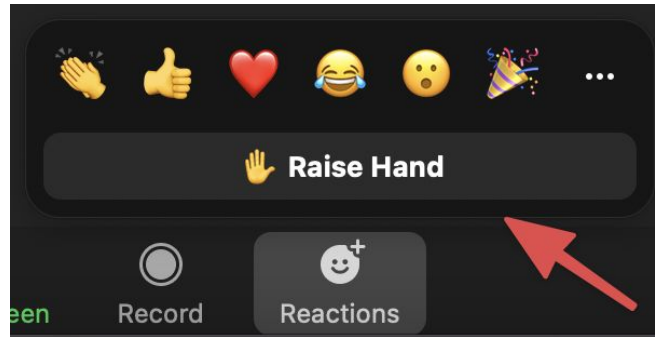
If you're able, please **turn on your camera!**
.... It can really make the section come to life!



(Image source: <https://as.virginia.edu/eight-ways-get-more-out-zoom>)

Zoom Reactions

- 👍 **Thumbs Up:** If you understand.
- 🙋 **Raise Hand:** If you have a question (or just speak in the mic).



Introductions

Hi, I'm David!

- Head TA, here at my 4th Code in Place.
 - Started as a CIP student! 2x volunteer Section Leader.
- CS @ Massachusetts Institute of Technology (MIT)
- Product Manager and former Software Engineer
- Love photography, video games, movies
- Guilty pleasure: Reality competition shows like Survivor
- Cool superpower I'd like to have:
 - Felix Felicis ("Liquid Luck" potion) from Harry Potter

How about you?

- Where are you from?
- Your coding experience
- Cool superpower you'd like to have

Course Weekly Timeline

Lectures

2 Lessons

Both released every Monday.
~1 hour per lecture.

Section

Live Zoom Session

Practice lecture concepts.
Prepare you for assignment.

Assignment

Coding Challenges

Solidify your understanding.
Optional: "Extension" projects

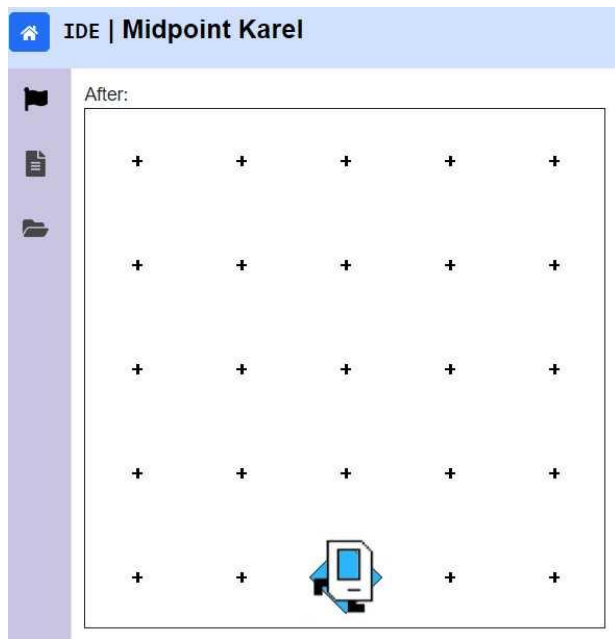
Brief Tour of Course's Web Platform

The screenshot displays the Code in Place web platform interface. The browser address bar shows the URL `codeinplace.stanford.edu/cip3/join/ide/a/piles`. The interface is divided into several panels:

- IDE | Piles**: The main header with a **Run** button and a **Share** button.
- Instructions**: Contains the task description: "Your task: Write a program in the editor, that makes Karel pick up all the beepers on the first row of this world." and a visual representation of the world with a 6x6 grid of beepers.
- main.py**: The code editor showing a Python program that uses the Karel API to pick up beepers on the first row of the world.
- World**: A visual representation of the world with a 6x6 grid of beepers. The first row contains 6 beepers, and the bottom row contains 3 beepers. A slider for "Text descriptions" is visible below the world.
- Terminal**: A panel at the bottom for running the program.

```
1 from karel.stanfordkarel import *
2
3 # File: piles.py
4 # -----
5 # The warmup program defines a "main"
6 # function which should make Karel
7 # pick up all the beepers in the world.
8 def main():
9     move()
10    pick_10()
11    move()
12    move()
13    pick_10()
14    move()
15    move()
16    pick_10()
17    move()
18
19 def pick_10():
20     for i in range(10):
21         pick_beeper()
22
23
24
25
26
27 # don't edit these next two lines
28 # they tell python to run your main function
29 if __name__ == '__main__':
30     main()
```

Some Basic Python Functionality is TURNED OFF....for now!

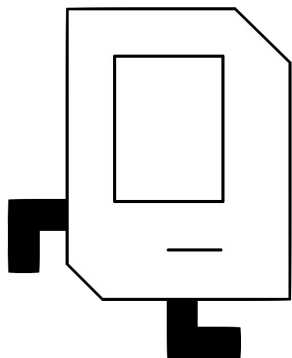


For the duration of Karel unit (Weeks 1 & 2):

- Basic Python functionality is turned off for simplicity
 - Focus on fundamentals of programming
- Examples of disabled functionality:
 - print statements
 - variables, lists, dictionaries
 - conditional keywords (**NOT**, etc.)
 - function return statements

Concepts Review (Lightning Round!)

Karel is like a LEGO set with only 4 types of blocks....but you can do a lot!



Karel only knows four commands at the beginning:

`move()`

`turn_left()`

`put_beeper()`

`pick_beeper()`

Helpful website to visualize Karel (by Sophia Westwood):

www.karelhelper.com

Control Flow



for **Loops**



while **Loops**



if/else

for Loop

```
for i in range(count):  
    statements                # note indenting
```

```
# Example:  
def turn_right():  
    for i in range(3):  
        turn_left()           # note indenting
```

while Loop

```
while condition:  
    statements                # note indenting
```

```
# Example:  
def move_to_wall():  
    while front_is_clear():  
        move()                # note indenting
```



Conditions Karel Can Check For

<i>Test</i>	<i>Opposite</i>	<i>What it checks</i>
<code>front_is_clear()</code>	<code>front_is_blocked()</code>	Is there a wall in front of Karel?
<code>left_is_clear()</code>	<code>left_is_blocked()</code>	Is there a wall to Karel's left?
<code>right_is_clear()</code>	<code>right_is_blocked()</code>	Is there a wall to Karel's right?
<code>beepers_present()</code>	<code>no_beepers_present()</code>	Are there beepers on this corner?
<code>beepers_in_bag()</code>	<code>no_beepers_in_bag()</code>	Any there beepers in Karel's bag?
<code>facing_north()</code>	<code>not_facing_north()</code>	Is Karel facing north?
<code>facing_east()</code>	<code>not_facing_east()</code>	Is Karel facing east?
<code>facing_south()</code>	<code>not_facing_south()</code>	Is Karel facing south?
<code>facing_west()</code>	<code>not_facing_west()</code>	Is Karel facing west?

Check out “Karel Reader”, Chapter 10 for full reference

Which Type of Loop Should You Use?

for Loop

You know exactly how many times to repeat (definite loop)

vs.

while Loop

You don't know exactly how many times to repeat (indefinite loop)

if-else Statement

```
if condition:  
    statements                # note indenting  
else:  
    statements
```

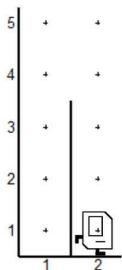
```
# Example:  
def invert_beepers():  
    if beepers_present():  
        pick_beeper()        # note indenting  
    else:  
        put_beeper()         # note indenting
```


Decomposition



Break down a problem into more manageable sub-problems

- “Divide and conquer” approach
- Which sorts of tasks should be decomposed?
- Rule of Thumb: Each function should execute a single purpose.
- Code becomes easier to read! For both you and everyone else.

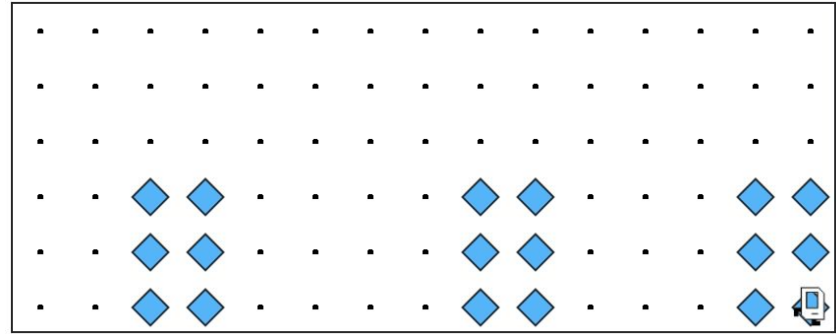
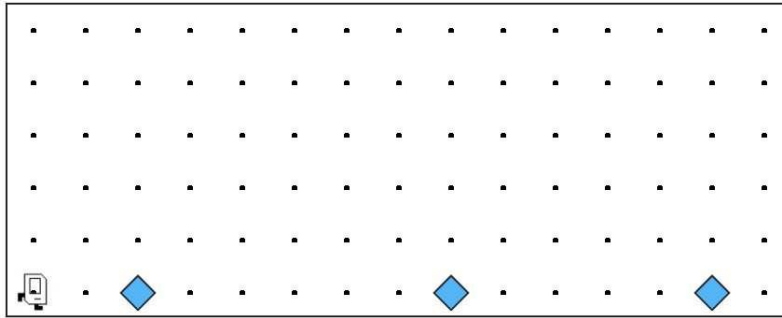


```
turn_left()
while right_is_blocked():
    move()
turn_right()
move()
turn_right()
move_to_wall()
turn_left()
```



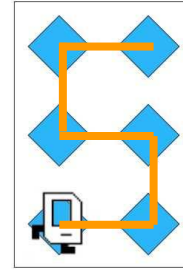
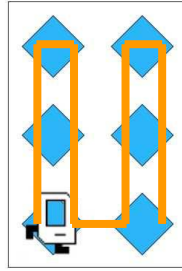
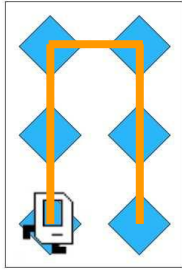
```
ascend_hurdle()
move()
descend_hurdle()
```

Section Exercise: "Hospital Karel"



Section Exercise: "Hospital Karel"

Which option would you pick to build the 2x3 hospital? Pros vs Cons?



What if the hospital was another shape?

